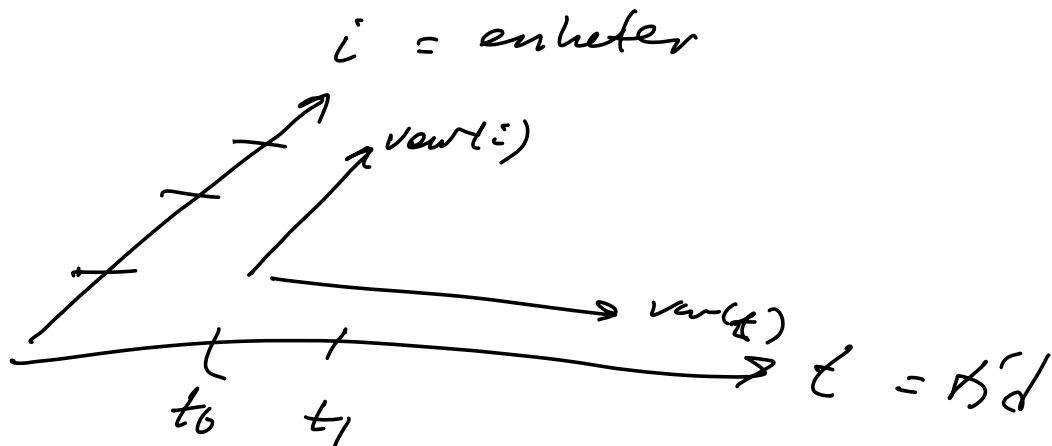


PANEL DATA ?



$$i = 1, \dots, N \quad t = 1, \dots, T$$

intercept $\downarrow$ TIME INVARIANT

$$y_{it} = \beta_1 + \beta_2 \cdot X_{it} + \underbrace{u_i + e_{it}}_{v_{it}} \quad \text{OLS (POOLED)}$$

$\downarrow$ idiosyncratic error

ONE	2	y	i	T	X	$\bar{X}$	$\bar{X}_2$
1	0	2	1	1	3	4	4.5
1	0	4	1	2	6	4	4.5
1	1	3	2	1	4	4	3.5
1	1	9	2	2	3	4	3.5
					$\Sigma 16/4 = 4$		

ENDOGENEITY
 $\text{cov}(X_{it}, v_{it}) \neq 0$

expenditure on liquor = $f(\text{household income})$

$$\text{cov}(u_i, X_{it}) = 0 ?$$

$$\text{corr}(v_{it}, v_{is}) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_e^2} = \rho$$

$$(X_{it} - \bar{X}) = (X_{it} - \bar{X}_i) + (\bar{X}_i - \bar{X})$$

TOTAL = within + BETWEEN

POOLABILITY TEST

$$y_{it} = \alpha + \beta X_{it} + v_{it} \quad (\text{POOLED OLS})$$

$$y_{it} = \alpha_i + \beta X_{it} + v_{it} \quad (\text{FIXED EFF})$$

↓

$$H_0: \alpha_1 = \alpha_2 = \dots = \alpha_{40} = 0$$

F-test